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1. Introduction

Organizations today have to manage human resources to the fullest extent to maintain employee productivity and decrease turnover. A significant challenge faced by organizations is the rate of employee attrition; this has an effect on the organization's performance, increases costs for hiring new employees, and disrupts the overall balance of the organization.

The focus of this project is to utilize Tableau to analyze data related to HR in order to determine the patterns and trends with relationships to employee attrition. The dataset used in this project contains specific attributes related to attrition such as department, job function, age range, and attrition status. The individual attributes allow for the identification of the factors that have contributed to employee retention.

The purpose of the project is to demonstrate how data visualization techniques can provide insight into how employee attrition varies based on employee segments. By developing a dashboard in Tableau, the data is represented in a visual and interactive format, which allows for easier identification of the trends within the data set.

This analysis will specifically highlight the areas where the rate of attrition is higher and investigate what the causes of the high rates of attrition are. This information will allow organizations to improve their decision-making processes for the purposes of enhancing employee satisfaction, improving retention strategies, and enhancing workforce management across their organization.

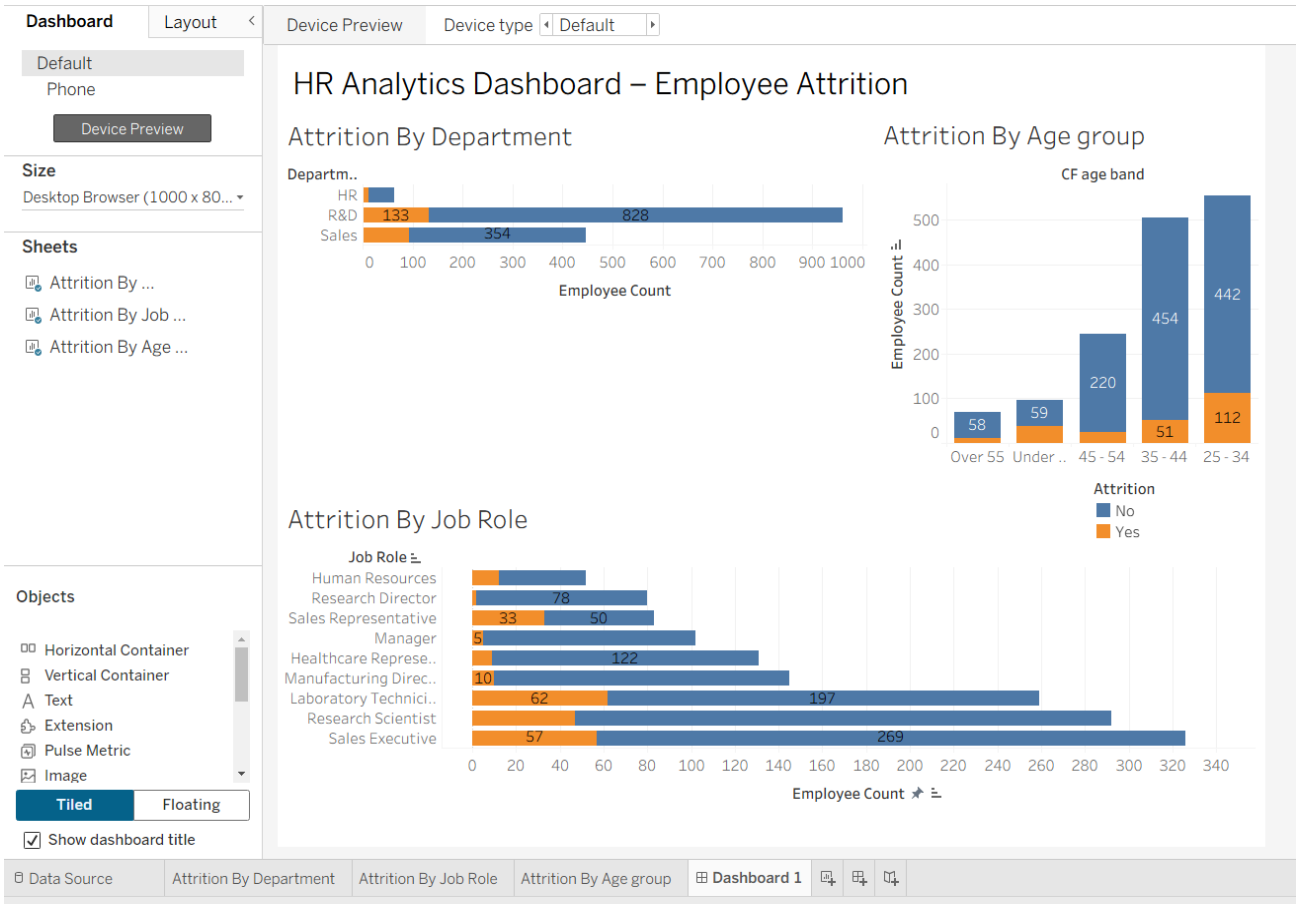
2. Dataset Discription

This project utilizes a dataset with data on all the employees of an organization in order to analyze characteristics of a company's workers along with why employees leave their jobs.

Key variables included in the dataset are department, job role, age group, the type of business travel involved, education type, and whether the employee has left the organization or is active (attrition).

The data set will provide an opportunity to identify relationships between different factors and employee turnover. The analyses will help identify patterns resulting in higher attrition to specific departments and job roles as well as how worker demographics influence retention.

3. Dashboard Output

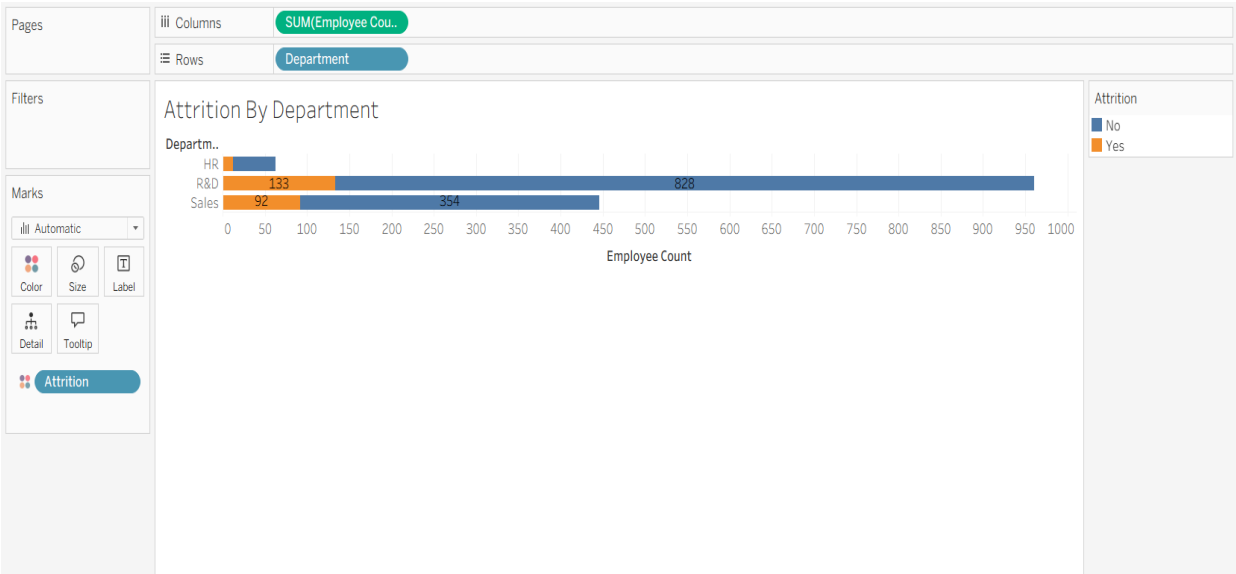


The above dashboard presents a comprehensive visualization of employee attrition using Tableau. It combines multiple charts to provide insights into how attrition varies across different dimensions such as department, job role, and age group.

The interactive nature of the dashboard allows users to easily explore the data and identify patterns. It helps in simplifying complex data into visual formats, making analysis more efficient and understandable.

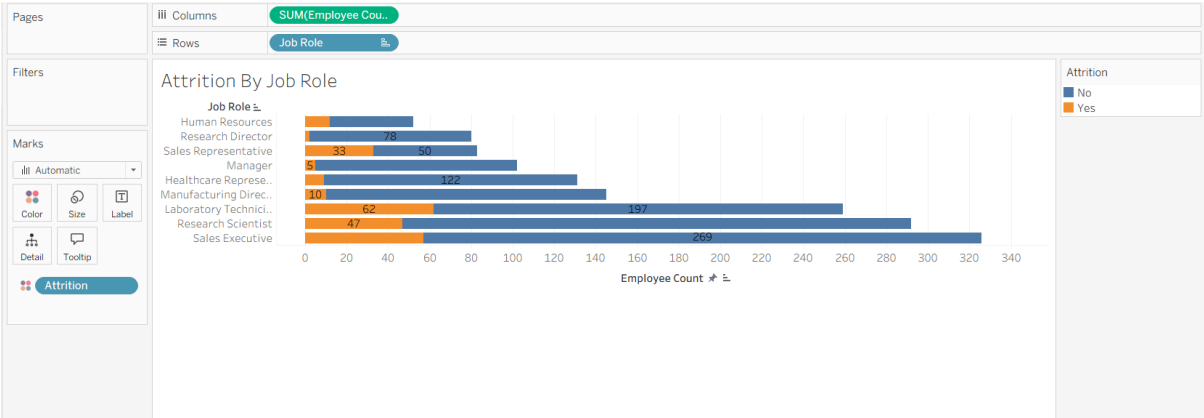
4. Charts Explanation

1)



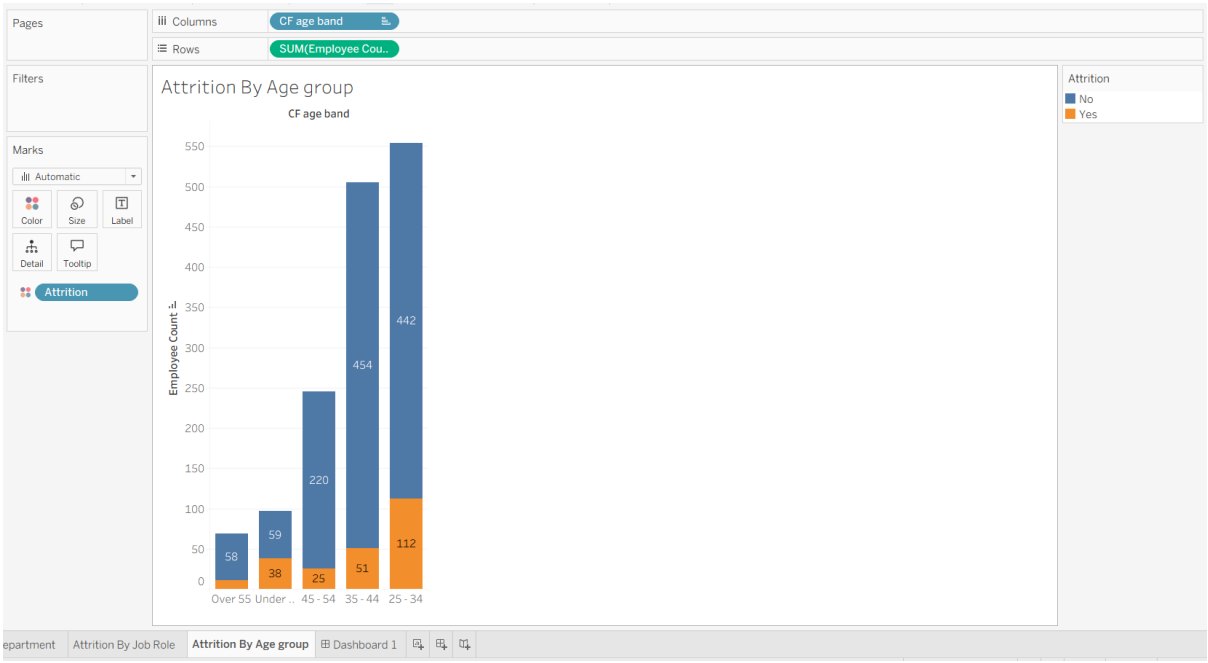
The "Attrition by Department" chart indicates the distribution of employee departures between jobs within various departments and highlights those with the highest level of retention.

2)



The "Attrition by Job Role" Chart Indicates the differences between job title and turnover rates and highlights those that have the highest level of employee departure.

3)



The "Attrition by Age" chart provides insight into the differences in rotability between ages of employees and demographic trends of employees staying employed.

5. Conclusion

The HR dataset was analyzed for insights regarding employee attrition. One of the major findings is that attrition varies across department and job role rather than being evenly distributed throughout the organization.

Certain job roles tend to experience higher turnover rates than others. This could indicate underlying issues, such as job satisfaction and/or excessive workload; younger employees tend to have higher rates of attrition than older employees because they may anticipate more opportunities for career advancement or a greater chance of moving to another employer.

The data in the dashboard gives an overall view of attrition and identifies the major areas in which organizations should concentrate their efforts to enhance their ability to retain employees. The analysis will provide HR departments with information to improve their decision making regarding employee retention and to develop successful strategies for retaining employees.